KEY TO MANTIS (INSECTA: MANTODEA) FAMILIES AND GENERA FROM NORTH AMERICA (INCLUDING MEXICO)

Translated from:

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1.- Antennae thickened at the base of the flagellum [fig. 1]; body elongated; wings reduced (brachypterous condition); the populations of the only North American species are represented entirely by females..**COPTOPTERYGIDAE:** ***Brunneria***
-.- Antennae long and filiform; body variable; with well-developed wings, brachypterous or even wingless (apterous condition); the specimens can be of both sexes...**2**

2.- Pronotum squared, as long as wide [fig. 2A]; length of the body equal or lesser than 20mm [fig. 2A]..**MANTOIDIDAE:** ***Mantoida***
-.- Pronotum longer than wide; length of the body greater than 20mm..**3**

3.- Anterior femora with 5 or more posteroventral spines...**4**
-.- Anterior femora with 4 or less posteroventral spines...**11**

4.- Well developed wings which resemble dead leaves, these have irregular edges and or distal lobes of variable size [fig. 2B]; if the anterior wings have straight borders, the body is small, green or colorful, and has a pronotum of less than 8mm in length [fig. 2C]..**5**
-.- Brachypterous or apterous condition; if the wings are well developed, these have straight edges and the body is usually large (pronotum greater than 10mm in length); in case it is a small species, coloration is grayish or yellowish brown..**7**

5.- Anterior wings resemble dead leaves, have irregular edges and or distal lobes of variable size [fig. 2B]...**ACANTHOPIDAE: 6**
-.- Anterior wings with straight borders [2C]....................................**ACONTISTIDAE:** ***Acontista***

6.- Vertex without ocellar tubercle [fig. 3A]...***Acanthops***
-.- Vertex with ocellar tubercle [fig. 3B]...***Pseudacanthops***

7.- Frons with a pair or tubercles [fig. 4A]; wings of the adults always well developed; anal area of the posterior wings with a blue spot in the middle of a yellowish field [fig. 4B]............................ ....**EREMIAPHILIDAE:** ***Iris***
-.- Frons smooth; wings variable, in case the wings are well developed, they do not have blue spots in the posterior wings..**8**

8.- Females bracypterous or apterous and males with variable wings; small species which live onto the soil; color grayish or yellowish brown..**AMELIDAE: 9**
-.- Both sexes with well-developed wings; large species which live amongst the vegetacion; color green...**PHOTINAIDAE: *Macromantis***

9.- Compound eyes pointy [fig. 5A]; males apterous or brachypterous...**10**
-.- Compound eyes rounded [fig. 5B]; males with well-developed wings.......................***Litaneutria***

10.- Both sexes apterous..***Yersiniops***
-.- Both sexes brachypterous...***Yersinia***

11.- Body dorsoventrally flattener [fig. 6A; fig. 7]; anterior wings mottled [fig. 6A]....................**12**
-.- Body elongated or cylindrical..**13**

12.- Margins of the prozona widened towards the head [fig. 6A]; discoidal spines arranged in zigzag [fig. 6B]; the abdomen of the females has no cuticular excrescences......................................
....**LITURGUSIDAE: *Liturgusa***
-.- Margins of the prozona parallel to the head [fig. 7]; discoidal spines arranged in a straight line; abdomen of the female with cuticular excrescences [fig. 7].........**EPAPHRODITIDAE: *Gonatista***

13.- Head with juxtaocular bulges very prominent [fig. 8A]; anterior tibiae with 1-4 (or 8) ventral spines and generally with 1-2 dorsal spines [fig. 8B]; only if the anterior tibiae has 8 ventral spines, it lacks dorsal spines...**THESPIDAE: 14**
-.- Head with juxtaocular bulges discreet; anterior tibiae with more than 4 ventral spines; without dorsal spines...**19**

14.- Anterior tibiae with 8 ventral spines and without dorsal spines.......................***Pseudomiopteryx***
-.- Anterior tibiae with 1-4 ventral spines and 1-2 dorsal spines..**15**

15.- Anterior femora with 4 discoidal spines...**16**
-.- Anterior femora with 3 discoidal spines...**18**

16.- Anterior tibiae with 3-6 posteroventral spines...***Pseudomusonia***
-.- Anterior tibiae with one posteroventral spine in dorsal position...**17**

17.- Pronotum short and thickened, with a metazona which measures a little more than the prozona [fig. 9A]; supracoxal dilatation located slightly ahead of the midpoint of the pronotum [fig. 9A]..***Oligonicella***
-.- Pronotum elongated with a metazona which measures twice more than the prozona [fig. 9B]; supracoxal dilatation located in the anterior third of the pronotum [fig. 9B].........................***Bistanta***

18.- Anterior tibiae with 2 strong ventral spines widely separated [fig. 10A]......................***Oligonyx***
-.- Anterior tibiae with 2 dorsal spines [fig 10B]..***Thesprotia***

19.- Anterior tibiae extremely reduced [fig. 11A], less than a third of the length of the femur and with few spines; body extremely slim; cerci widely flattened [fig. 11B, 11C]..**ANGELIDAE:** ***Angela***
-.- Anterior tibiae not as reduced and generally with numerous spines; body variable; cerci cylindrical...**20**

20.- Tibial spur groove located in the proximal half of the anterior femora [fig. 12]; anterior legs with coxae which have 5-6 black spots and femora with have 3 black spots [fig. 12]..**MIOMANTIDAE:** ***Miomantis***
-.- Tibial spur groove located in the distal half, almost at midpoint of the anterior femora; color pattern of the anterior legs otherwise...**MANTIDAE: 21**

21.- Pronotum with extremely widened lateral projections [fig. 13A, 13B]..................***Choeradodis***
-.- Pronotum without said projections...**22**

22.- Head with two notorious ocellar processes, similar to horns [14A]......................................**23**
-.- Head usually without ocellar processes, if it has then, it is just one..**24**

23.- Lateral margins of the pronotum without expansion [fig. 14B]; dorsal surface of the pronotum with granules or small tubercles [fig. 14]; middle femora and tibiae sinuous [fig. 14C]...***Pseudovates***
-.- Lateral margins of the pronotum expanded around the supracoxal dilatation [fig. 15A]; surface of the pronotum smooth [fig. 15A]; middle femora and tibiae straight [fig. 15B]......................***Vates***

24.- Males with a middle ocellar process [fig. 16A]; females without said process, with an enlarged abdomen, almost as wide as large [fig. 16B]; anterior wings of the female with a spot in the center of the discoidal area [fig. 16B]...***Hondurantemna***
-.- Males without ocellar process; females with an abdomen clearly larger than wide; anterior wings of the female variable...**25**

25.- Anterior tibiae less than half the length of the anterior femora; inner surface of the anterior femora with black, white and/or red spots [fig. 17]..***Statilia***
-.- Anterior tibiae more than half the length of the anterior femora; inner surface of the anterior femora lacking spots...**26**

26.- Frons twice taller than wide, or less [fig. 18A]; anterior wings of the female always reach the tip of the abdomen..**27**
-.- Frons more than twice taller than wide [fig. 18B]; anterior wings of the female are shorter than the abdomen...**28**

27.- Inner base of the anterior coxae without spots; posterior wings lightly to strongly mottled..***Tenodera***
-.- Inner base of the anterior coxae with a large dark spot [fig. 19]; posterior wings completely hyaline...***Mantis***

28.- Pronotum with two tubercles in the anterior portion of the metazona [fig. 20].............***Melliera***
-.- Pronotum without said tubercles..**29**

29.- Anterior wings of the female cover at least half of the abdomen [fig. 21A]; posterior wings of both sexes with a violet spot in the anal area [fig. 21A]; cerci large and compressed [fig. 21B]; length of the body greater than 60mm [fig. 21A]...***Phasmomantis***
-.- Anterior wings of the female cover around three fourths of the abdomen; posterior wings of both sexes usually hyaline [fig. 22A], although they can be pigmented [fig. 22B]; cerci normal; length of the body rarely greater than 60mm..***Stagmomantis***